Data Science Introduction (Demo PDF)

Data science is the practice of turning raw data into useful insights for decisions and automation.
It combines statistics, domain knowledge, and programming to solve practical business and research problems.

Typical lifecycle: define problem -> collect data -> clean/prepare -> explore -> model -> evaluate -> deploy -> monitor.

Core skills: Python/SQL, probability and statistics, machine learning basics, visualization, communication, and experimentation.

Common tasks: churn prediction, demand forecasting, recommendation systems, anomaly detection, NLP, and dashboards.

Key tools: pandas, NumPy, scikit-learn, matplotlib/seaborn, Jupyter, dbt, Airflow, and cloud platforms.

Good practices: clear problem framing, reproducible pipelines, train/validation/test splits, baseline models, error analysis, and model monitoring.

Ethics and governance: privacy, fairness, bias mitigation, explainability, and responsible AI usage are essential.

Career paths: data analyst, data scientist, ML engineer, analytics engineer, and data product manager.

Mini glossary:

- Feature: an input variable used by a model
- Label/Target: the value to predict
- Overfitting: model learns noise, not general patterns
- Precision/Recall: metrics for classification quality